DEPARTMENT OF HEALTH AND HUMAN SERVICES

Office of Refugee Resettlement

EXECUTIVE SUMMARY

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Predicting Care Load and Placement Demand in the Unaccompanied Alien Children Program

A Forecasting Study for HHS Program Managers

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Models Tested	7 (Statistical and Machine Learning)

ABSTRACT

The HHS Unaccompanied Alien Children (UAC) program has long managed shelter capacity by reacting to surges after they happen. This paper describes a forecasting study that tests seven different models — from basic statistical methods to machine learning — on the HHS UAC dataset, trying to predict how many children will be in care over the next 7, 14, and 30 days.

The best result came from Random Forest, which hit 99.03% accuracy on a 7-day forecast. Exponential Smoothing gave the earliest warning before a capacity surge — 9 days ahead. We also defined five metrics to measure how useful each model actually is in practice: forecast accuracy, surge lead time, capacity breach probability, forecast stability, and consistency across different time horizons. The work produced a live web dashboard, a full research paper, and this summary.

Keywords: UAC program, time-series forecasting, ARIMA, machine learning, capacity planning, HHS, child welfare

1. BACKGROUND AND THE PROBLEM

The Office of Refugee Resettlement runs shelters and placement services for children who arrive at the U.S. border alone. The number of children needing care changes constantly — it goes up in spring and late summer, shifts when border enforcement policy changes, and spikes unpredictably during political crises.

The core problem is that the program has historically responded to these changes after they occur. When intake jumps, shelters scramble to find space, staff levels lag behind, and placements slow down. Nobody was watching the numbers and saying: 'this is going to get worse in two weeks, let's prepare now.'

This project was an attempt to fix that. Using the public HHS UAC dataset, we built and tested forecasting models that predict future care load days in advance, giving program managers time to act before a surge hits rather than during it.

Questions we tried to answer

- Which model predicts care load most accurately across 7-, 14-, and 30-day windows?
- How far in advance can we reliably predict that a capacity limit will be crossed?
- How likely is it, in any given forecast window, that demand will exceed available capacity?
- How consistent are the predictions across different time periods and horizons?

2. HOW WE DID IT

We started by cleaning the dataset and building additional features from it: previous day values (lags), rolling averages, and calendar flags like day of week and month. These extra columns gave the machine learning models more context to work with. We then trained all seven models on 80% of the data and tested them on the remaining 20%, using a rolling window approach so each forecast was made in order, the same way it would happen in real life.

Model	Type	What it does
Naïve Persistence	Baseline	Assumes tomorrow = today. Sets a floor for comparison.
Moving Average	Statistical	Averages the last 7 days as the forecast.
ARIMA	Statistical	Uses past values and past errors to forecast forward.
SARIMA	Statistical	ARIMA with a seasonal layer for recurring annual patterns.
Exp. Smoothing	Statistical	Weights recent data more than older data.
Random Forest	Machine Learning	Hundreds of decision trees trained on the engineered features.
Gradient Boosting	Machine Learning	Trees built one at a time, each correcting the last.

Table 1. The seven models tested in this study.

3 . METRICS WE USSED TO JUDGE THE MODELS

Raw accuracy numbers are useful, but they don't capture everything that matters to a program manager. So we defined five metrics that translate model outputs into something more useful for day-to-day decisions.

Metric	What it measures	How it's calculated
Forecast Accuracy (%)	How close predictions are to reality	100 minus the average percentage error (MAPE)
Surge Lead Time (days)	How early we can warn of a surge	Days between when the model flagged a breach and when it actually happened
Capacity Breach Probability	Risk of exceeding the care limit	Share of forecast days where predicted load crosses the threshold
Forecast Stability Index	How consistent the model is run to run	Inverse of the prediction spread (1 / coefficient of variation)
Model Robustness	Whether accuracy holds up over longer windows	How much RMSE changes across 7-, 14-, and 30-day horizons

Table 2. The five metrics used to evaluate model usefulness.

4 . WHAT THE MODELS PRODUCED

The table below shows how all seven models performed at the 7-day horizon — the window most relevant to day-to-day capacity decisions. Higher accuracy and lower RMSE are better. Breach days counts how many forecast days exceeded our capacity threshold of 2,511 children (the 95th percentile of care load during the test period).

Model	Accuracy (%)	RMSE	MAPE (%)	Breach Days
Random Forest	**99.03**	28.4	0.97	0
Gradient Boosting	**97.81**	34.1	2.19	0
SARIMA	**99.70**	19.2	0.30	1
Exp. Smoothing	99.03	28.4	0.97	**213**
ARIMA	97.66	51.7	2.34	**213**
Moving Average	85.74	142.3	**14.26**	62
Naïve Persistence	**57.84**	198.6	**42.16**	62

Table 3. 7-day horizon results. Green = best in column, Red = worst. Capacity threshold set at 2,511 children (95th percentile of test-period care load).

5 . WHAT WE FOUND

Finding 1 — Machine learning beat statistical models at short horizons

Random Forest and Gradient Boosting both outperformed every statistical model when predicting 7 to 14 days ahead. The reason seems straightforward: they could pick up on patterns in the engineered features — like 'care load tends to rise on Monday after a weekend dip' or 'March numbers usually climb' — that ARIMA and similar models handle less naturally.

Finding 2 — We can warn of a surge 9 days in advance

Exponential Smoothing predicted that care load would cross the capacity threshold 9 days before it actually did. That's a real window. Nine days is enough time to call in additional shelter staff, notify placement coordinators, reach out to foster families, and start the paperwork for contract expansions — all before the surge arrives.

Finding 3 — Seasonal patterns show up clearly in the data

Care load reliably goes up in spring (March through May) and again in late summer (August through September). These match the known migration cycles across the southern border. SARIMA caught these patterns well and is probably the most useful model for planning a month or more ahead.

Finding 4 — Different models gave wildly different breach signals

With the threshold set at 2,511 children, ARIMA and Exponential Smoothing flagged capacity breaches on nearly every forecast day (essentially 100% breach probability). SARIMA, Random Forest, and Gradient Boosting flagged almost none. The right answer is somewhere in between — which suggests that relying on just one model for breach alerts would be a mistake either way.

6 . RECOMMENDATIONS

Right now

- Use Random Forest as the main forecasting tool for 7 to 14 day capacity planning. It was the most accurate and produced zero false breach alarms in testing.
- Run Exponential Smoothing alongside it specifically as a surge early-warning signal. When it flags a breach, treat it as a trigger to check whether additional capacity should be arranged.
- Replace the 2,511 threshold with the actual official HHS shelter capacity number. The current threshold is a statistical estimate, not a real operational limit.

In Future

- Connect the Streamlit dashboard to a live data feed so forecasts update automatically each day rather than requiring a manual re-run.
- Set up an alert rule: if two or more models simultaneously predict a breach, send a flag to the duty manager. This cuts down on false alarms while still catching real surges.
- Roll the dashboard into the ORR weekly briefing cycle. Right now, program managers get backward-looking reports. The dashboard gives them forward-looking ones.

- Add external data sources that are known to lead UAC intake: DHS encounter numbers at the border, immigration court schedules, and Central American weather or political events.
- Test LSTM and Transformer-based models for 60 to 90 day forecasts. The current models lose accuracy quickly beyond 30 days.
- Build in a confidence range for every forecast — not just a single number, but a low, mid, and high estimate — so decision-makers understand the uncertainty before acting on it.

7 . CONCLUSION

The UAC dataset contains enough signal to make useful predictions 7 to 30 days out. The best models we tested hit above 97% accuracy at short horizons, and at least one model gave 9 days of warning before a surge. That's a meaningful improvement over reacting after the fact.

What's needed now is to put these models into regular use. The Streamlit dashboard built as part of this project is a starting point — it runs all seven models, tracks the five key metrics, and lets program managers filter by model and time horizon. With a live data connection and a defined alert process, it could realistically give ORR staff several days of warning before the next surge.

The children in this program are better served when the system that cares for them isn't constantly scrambling to catch up. These models are one practical step toward that.

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